

SMS Spam Classification Using Artificial Neural Networks

1. Summary of Approach

This project developed an SMS spam classification system using an Artificial Neural Network (ANN). The dataset contained **5,572 SMS messages**, consisting of **4,825 ham messages (86.59%)** and **747 spam messages (13.41%)**.

The text messages were preprocessed by converting text to lowercase, removing URLs, punctuation, digits, and special characters. The cleaned text was then tokenized, stopwords were removed, and the remaining words were lemmatized.

After preprocessing, the SMS messages were converted into numerical features using **TF-IDF (Term Frequency–Inverse Document Frequency)** vectorization. The dataset was divided into training and testing sets. The TF-IDF features were used to train a feedforward ANN consisting of hidden layers with ReLU activation and an output layer with Sigmoid activation for binary classification.

Early stopping was used during training to reduce overfitting.

2. Key Results

The ANN was evaluated using Accuracy, Precision, Recall, and F1-score. Logistic Regression was also trained as a baseline model using the same TF-IDF features and train-test split.

Model	Accuracy	Precision	Recall	F1-Score
ANN	97.93%	95.65%	88.59%	91.98%
Logistic Regression	96.95%	97.52%	79.19%	87.4%

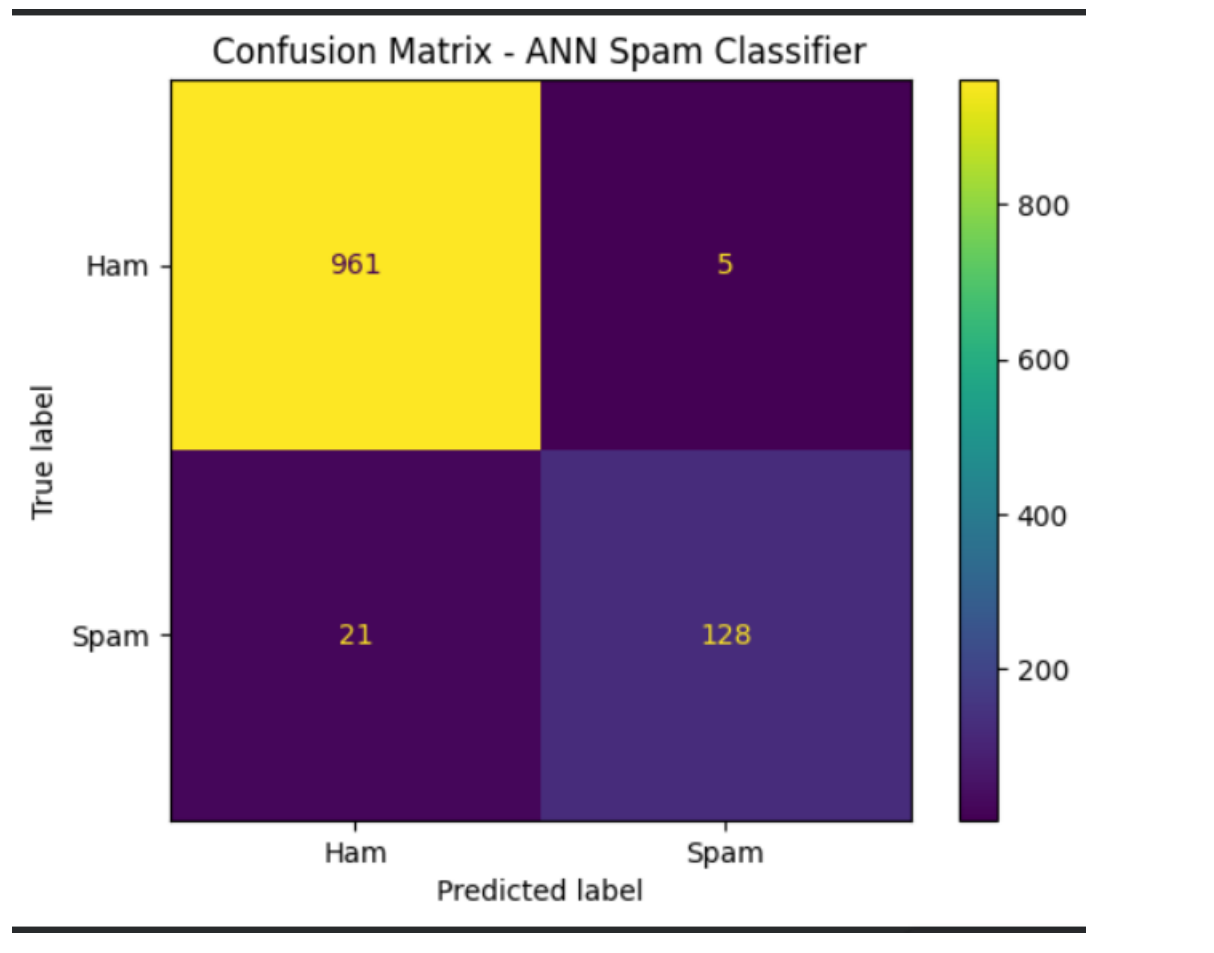
The ANN achieved better overall performance in terms of Accuracy, Recall, and F1-score. The higher Recall indicates that the ANN successfully detected more actual spam messages. Logistic Regression achieved higher Precision, meaning that when it predicted spam, it was less likely to incorrectly classify a ham message as spam.

3. Confusion Matrix and Error Analysis

The confusion matrix was used to analyze the model's classification performance by identifying correct and incorrect predictions.

The error analysis examined five False Positives and five False Negatives. False Positives occurred when legitimate ham messages were incorrectly classified as spam. These errors were mainly associated with very short messages, unusual word combinations, and messages with probabilities close to the 0.5 classification threshold.

False Negatives occurred when spam messages were incorrectly classified as ham. Several of these messages used conversational or informal language and lacked typical spam-related keywords. This demonstrates a limitation of TF-IDF: it represents word importance but does not fully capture a message's semantic meaning and context.



4. Reflections, Limitations, and Improvements

The ANN improved the overall performance compared with the Logistic Regression baseline, particularly in Recall and F1-score. However, this improvement involves several trade-offs.

One limitation is that the ANN is less interpretable than Logistic Regression. Logistic Regression allows individual feature coefficients to be examined, making it easier to understand which words influence predictions. In contrast, the ANN operates through multiple neurons and hidden layers, making its decisions more difficult to explain.

Another limitation is training complexity. Logistic Regression trains faster and works efficiently with sparse TF-IDF data, whereas the ANN requires more computational resources and hyperparameter tuning.

The dataset is also imbalanced, with only 13.41% spam messages. Future improvements could include using class weights, oversampling techniques, hyperparameter tuning, and more advanced NLP models such as word embeddings, LSTM networks, or transformer-based models.

Overall, the ANN provided the best performance in this experiment, achieving an **F1-score of 91.98%** and demonstrating stronger spam detection capability than the baseline model.